

JOLAN THELAN GOVENDER

BScHons Actuarial Science Candidate | University of the Witwatersrand

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PROFESSIONAL SUMMARY

Final-year BScHons Actuarial Science student at Wits with 8 ASSA exemptions and a research focus on machine learning applied to mortality forecasting. Drawn to problems where data and real-world consequences meet, which is what pulled me toward actuarial science and keeps me reading about financial markets and economic history in my own time. Outside of coursework, I tutor over 120 second-year students, which has made me as comfortable explaining a concept clearly as I am deriving it and strengthened my communication skills.

Outside of academics, I prioritise staying active, whether on the golf course, padel court, or a hiking trail. I strongly believe a balanced lifestyle is key to maintaining mental clarity and professional excellence, keeping me grounded, focused, and energised in my work. Beyond sports, I enjoy travelling globally to experience different cultures and cuisines.

EDUCATION

University of the Witwatersrand

BScHons – Actuarial Science (2026)

2026 – Present

- Actuarial Risk Management (A311)
- Life Insurance (F102)
- Health, Social and Employee Benefits (F108)

BSc – Actuarial Science & Mathematical Statistics

2023 – 2025

Star College Boys High School, Durban

National Senior Certificate – 7 Distinctions, 88% Average

2018 – 2022

ASSA EXEMPTIONS

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|---|---|
| • A112 (Business Economics) | • A213 (Contingencies) |
| • A113 (Business Finance) | • A214 (Actuarial Economic Modelling) |
| • A211 (Financial Mathematics) | • N111 (Foundation Actuarial Professional Practice) |
| • A212 (Risk Modelling & Survival Analysis) | • N211 (Communications) |

RESEARCH

Mortality Forecasting Using Recurrent Neural Networks

BScHons Research Project · University of the Witwatersrand

2026 – Present

Constructed and evaluated Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) recurrent neural network architectures for long-horizon mortality forecasting, benchmarked against the classical Lee-Carter model on a 10-year test set.

Key findings: Both neural architectures outperformed Lee-Carter, which showed systematic over-prediction at pension ages in both populations, consistent with its single-index constraint; however, neither beat a naive random walk with drift, and liabilities varied by under 1% across all models.

Findings speak directly to concerns raised in actuarial literature around the reliability and stability of neural network approaches to mortality projection.

EXPERIENCE

Tutor – Actuarial Science (2nd Year)

University of the Witwatersrand

2026 – Present

- Support 120+ second-year students across Financial Mathematics, Business Applications, and Life Contingencies through weekly tutorials.

MiWay Actuarial Hackathon

MiWay Insurance

2026

- Modelled the risk cost and office premium for 500 000 policies, simulating multiple years in a competitive insurance market, and placed 3rd out of the competing teams.
- Created Light Gradient Boosting-Machine models, split into frequency and severity, to accurately capture the interactions between different rating factors.

TECHNICAL SKILLS

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|------------------------------|--|-----------------------------------|
| • R (Advanced) | • Life Contingencies | • Pricing and experience analysis |
| • Microsoft Excel (Advanced) | • Mortality/longevity modelling | • Frequency-severity modelling |
| • Financial Mathematics | • Statistical analysis | • Rating factor analysis |
| • Risk Theory & Reserving | • Machine Learning (RNNs, LSTM, GRU, GBMs) | |

LANGUAGES

- English (Native) | Afrikaans (Working Proficiency)

AWARDS & INTERESTS

- Good Fellowship Award – Star College Boys High, 2022
- Bronze Medal – Golden Climate Environmental Project Olympiad
- 3rd Place – MiWay Actuarial Hackathon 2026
- Interests: Golf, Padel, Weight-lifting, Financial Markets, Chess

REFERENCES

Rivaj Maharaj (FASSA)

Lecturer, University of the Witwatersrand

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Rendani Mbuva (FASSA, CERA)

Associate Professor, Actuarial Science, University of the Witwatersrand

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